Akoizo Original Practice - Form 2 - Answer key

Division C | Protein Modeling | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Translation builds a chain from…

Unit: Protein synthesis and diverse functions.
Worked-case data: Coding DNA is 5′-ATG GAA TAA-3′. Give simple unprocessed mRNA and translated product.

A. Both ends randomly

B. DNA ends directly

C. N terminus toward C terminus

D. C toward N

Correct: C

2. (2 points) Attachment and fusion are…

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

A. Only DNA processes

B. Absent in all enveloped viruses

C. Different entry-related steps

D. Always identical

Correct: C

3. (2 points) A strong annotation explains…

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

A. Only a teacher name

B. Only material cost

C. Identity, role, and evidence

D. Only a color

Correct: C

4. (2 points) Quaternary structure requires…

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

A. No interfaces

B. Multiple polypeptide subunits

C. Exactly four atoms

D. Only one residue

Correct: B

5. (2 points) Reassortment exchanges…

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

A. Only antibodies

B. Genome segments

C. Only individual atoms

D. Only amino-acid colors

Correct: B

6. (2 points) A model's first design decision is…

Unit: Event evidence, reference design, and model purpose.
Worked-case data: A structure entry has chains A and B and a gap at residues 40-45. Plan a model label.

A. Its scientific purpose

B. Its price

C. Its teacher name

D. Maximum decoration

Correct: A

7. (2 points) More geometric contacts prove greater stability?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

A. No

B. Always

C. Only in cartoons

D. Only for enzymes

Correct: A

8. (2 points) The built-in lab executes full Jmol?

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain's helix region.

A. No

B. Yes

C. Only at maximum zoom

D. Only for HA

Correct: A

9. (4 points) Why label 5′ and 3′?

answer: Orientation determines the correct complement and reading frame. (4 points).

10. (4 points) Name an innate mechanism.

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

answer: Pattern recognition or interferon signaling, among other defenses. (4 points).

11. (4 points) Why label both termini?

Unit: Pre-build planning, annotation, and model review.
Worked-case data: At a scale of 1 cm per Å, how long is a modeled 12 Å separation?

answer: They establish directionality and help trace the chain. (4 points).

12. (4 points) Why inspect assembly annotations?

Unit: Structural levels and denaturation.
Worked-case data: Heating disrupts activity but the amino-acid sequence remains intact. What changed?

answer: Crystal neighbors may not be biological partners. (4 points).

13. (4 points) Why inspect glycans?

Unit: Influenza H1N1 and hemagglutinin.
Worked-case data: A model shows one HA chain. What must be checked before describing the functional surface?

answer: They can affect accessibility and recognition and may be incompletely represented. (4 points).

14. (4 points) Why use a legend?

Unit: Event evidence, reference design, and model purpose.
Worked-case data: A structure entry has chains A and B and a gap at residues 40-45. Plan a model label.

answer: Colors and symbols have no universal meaning without definitions. (4 points).

15. (4 points) Why are membrane proteins different?

Unit: Amino-acid chemistry, folding, and stability.
Worked-case data: A buried leucine is replaced by a charged residue. Predict cautiously.

answer: Their nonpolar surfaces can interact favorably with lipid rather than bulk water. (4 points).

16. (4 points) What makes a workflow reproducible?

Unit: Molecular viewing and reproducible JUDE/Jmol workflows.
Worked-case data: Plan a figure highlighting one chain’s helix region.

answer: Documented input data and ordered operations with consistent identifiers. (4 points).

17. (4 points) How would you test a functional substitution?

Unit: Protein synthesis and diverse functions.
Worked-case data: Coding DNA is 5′-ATG GAA TAA-3′. Give simple unprocessed mRNA and translated product.

answer: Compare a controlled mutant and reference in an appropriate binding or activity assay. (4 points).

18. (4 points) Why not assign epitopes from the synthetic lab?

Unit: Viruses, entry, and immune recognition.
Worked-case data: An antibody binds a viral protein in an assay. Does that prove neutralization?

answer: It has no experimental viral sequence or validated interaction annotations. (4 points).